

Practice Quiz: Planning / Ethics (Philosophical Foundations)

Name: __ Date: __

Part A: Multiple Choice

1. Hume's is-ought problem is relevant to Active Inference because: A) Active Inference solves it completely B) The FEP describes what organisms *do* (minimize free energy) but cannot directly prescribe what they *should* do C) Active Inference ignores values D) Ethics is irrelevant to science
2. The C vector in Active Inference represents: A) The agent's color preferences B) The agent's preferred observations — the formal representation of values and goals C) The complexity of the model D) The number of states
3. The hedonic-eudaimonic distinction connects to Active Inference because: A) Free energy minimization only supports hedonism B) EFE's epistemic term drives curiosity and growth, suggesting that genuine free energy minimization over long horizons is eudaimonic, not merely hedonic C) Both are about pleasure D) Neither applies
4. The AI alignment problem in Active Inference terms is: A) Making AI fast B) Setting the AI's C vector (values) to align with human values — and ensuring learned preferences stay aligned C) Aligning code formatting D) Physical alignment of hardware
5. Active Inference's ethical advantage for AI is: A) It's cheaper B) Interpretability — you can inspect the A, B, C, D matrices and understand *why* the agent acts as it does C) It uses less energy D) It's always correct
6. Moral emotions like guilt are understood in Active Inference as: A) Irrational B) Prediction errors within the social-moral generative model — guilt arises when actions violate one's own moral priors C) Random noise D) External punishments

7. Political power as "precision control" means: A) Power is about physical force B) Power = the ability to control what information is attended to — to set the precision of others' prediction errors C) Power is about money D) Precision is irrelevant to politics

Part B: Essay Questions

1. "Active Inference shows that to be alive is to have values." Evaluate this claim. Does the fact that all living systems maintain homeostatic setpoints (preferred states) constitute genuine "valuing," or is it merely a metaphorical extension of the concept? Does this bridge the is-ought gap? Reference at least two philosophical positions on the nature of values. (400 words)

2. You are tasked with designing an AI Active Inference system for healthcare. Specify: (a) How would you set the C vector to encode medical ethics principles (beneficence, non-maleficence, autonomy, justice)? (b) How would you handle conflicts between these principles? (c) What safeguards would prevent value drift through learning? (d) Should the system have any autonomy, or should it function purely as a decision-support tool? Justify each choice. (400 words)

3. Reflect on the entire Philosophical Foundations track. Among the eight philosophical dimensions we've explored (epistemology, philosophy of mind, phenomenology, 4E cognition, pragmatism, philosophy of science, social philosophy, ethics), which provides the most compelling philosophical case for Active Inference, and which presents the greatest challenge? Write a synoptic review that ties together at least four modules. (500 words)